

Assignment Week 3B – Biological Database

Based on this week's lessons on biological databases, complete the following task and submit your answers by pasting them in the space provided.

Using the TCGA data portal:

1. Search Criteria

1. **Disease:** Liver cancer
2. **Primary site:** *Adenocarcinoma, NOS*
3. **Experimental strategy:** *RNA-Seq*

2. Using your search results, answer the following:

- a. How many **mutated genes** are displayed?
- b. How many **total mutations** are present?
- c. Which gene has the **highest number of mutations**?
- d. What is the **mutation frequency** of this gene in liver cancer (in percentage)?
- e. What is the **amino acid change** and at what **position** did it occur?
(e.g., *Alanine (A) → Arginine (R) at position 175*)

Biological Impact (Advanced Task - Optional)

1. Using TCGA's '**Impact**' column (VEP, PolyPhen or SIFT):
 - f. What is the predicted effect of this mutation? (e.g., "probably damaging", "deleterious", "moderate impact")
 - g. How might this mutation affect the protein's structure or function? (Briefly explain in 2–3 sentences.)

ANSWERS

The screenshot shows the GDC Analysis Center Cohort Builder interface. The top navigation bar includes the NIH logo, GDC Data Portal, and various utility links like Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart, Login, and GDC Apps. The main header shows the current cohort name 'Unsaved_Cohort' and a status 'Cohort not saved'. Below this, there are filters for Disease Type (adenomas and adenocarcinomas), Primary Site (liver and intrahepatic), and Experimental Strategy (RNA-Seq). The main content area is titled 'COHORT BUILDER' and contains several tables for selecting and viewing cohort members.

General

Program

Name	Cases
☐ HCMI	5 (0.01%)
☐ MATCH	19 (0.04%)
☐ TCGA	403 (0.80%)

Project

Name	Cases
☐ HCMI-CMDC	5 (0.01%)
☐ MATCH-B	1 (0.00%)
☐ MATCH-C1	1 (0.00%)
☐ MATCH-H	2 (0.00%)
☐ MATCH-I	3 (0.01%)
☐ MATCH-P	1 (0.00%)

Disease Type

Name	Cases
☒ adenomas and adenocarcinomas	427 (0.85%)
☐ complex mixed and stromal neoplasms	2 (0.00%)
☐ mature t- and nk-cell lymphomas	1 (0.00%)
☐ neoplasms, nos	2 (0.00%)
☐ neuroepitheliomatous neoplasms	2 (0.00%)
☐ nevi and melanomas	1 (0.00%)

Primary Diagnosis

Name	Cases
☐ adenocarcinoma, nos	13 (0.03%)
☐ basal cell carcinoma, nos	5 (0.01%)
☐ basaloid squamous cell carcinoma	2 (0.00%)
☐ cholangiocarcinoma	54 (0.11%)

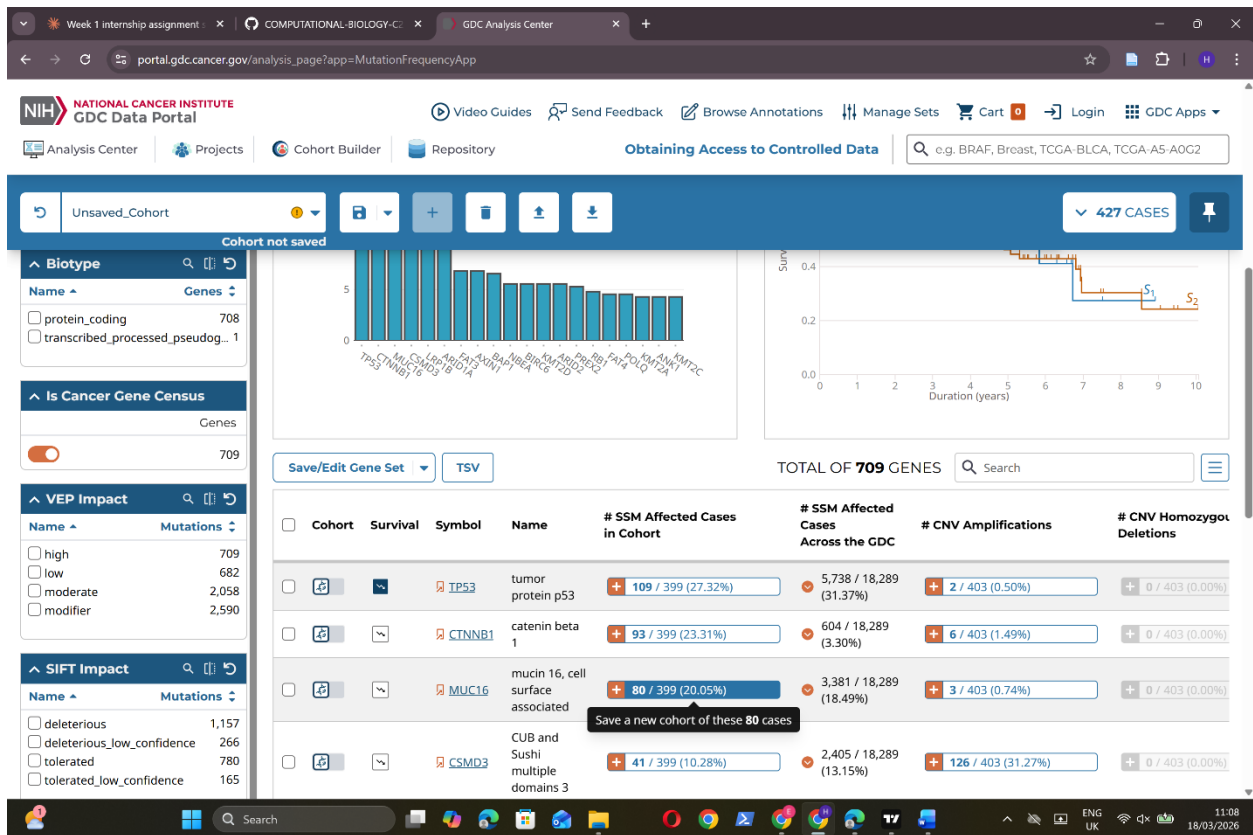
Primary Site

Name	Cases
☐ adrenal gland	80 (0.16%)
☐ anus and anal canal	1 (0.00%)
☐ bladder	2 (0.00%)
☐ breast	5 (0.01%)

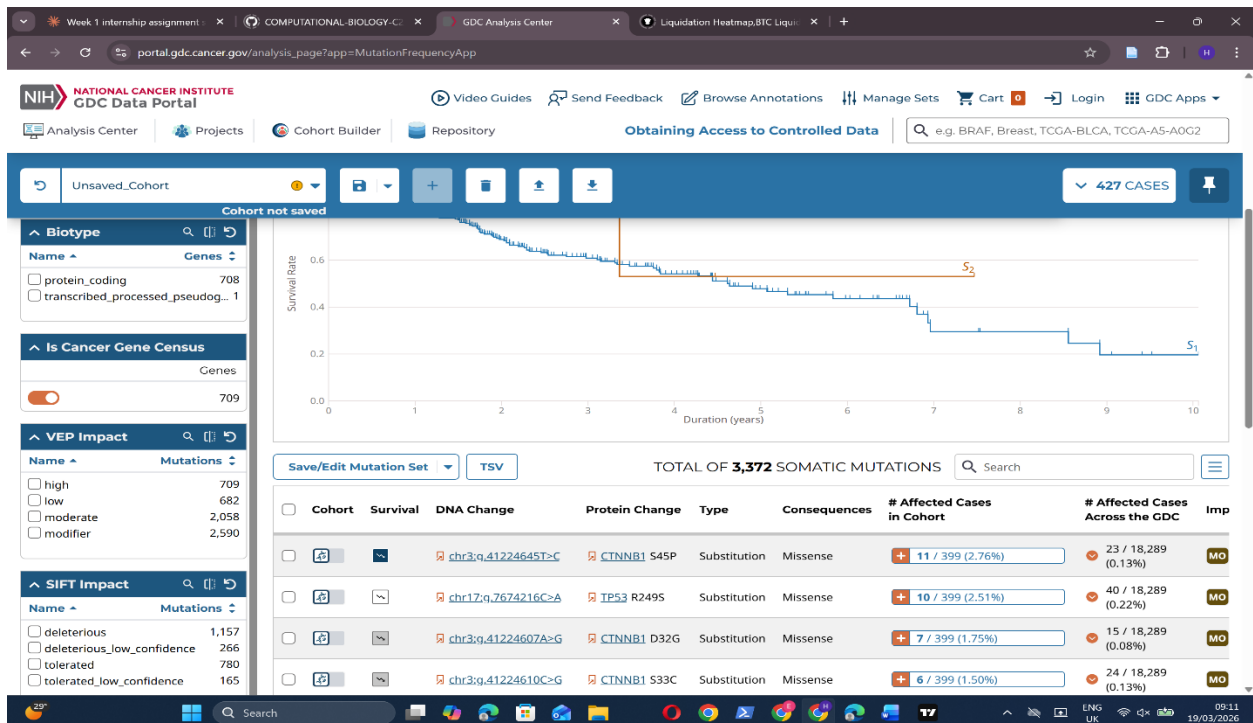
Tissue or Organ of Origin

Name	Cases
☐ abdomen, nos	2 (0.00%)
☐ adrenal gland, nos	4 (0.01%)
☐ bladder, nos	3 (0.01%)
☐ bone, nos	11 (0.02%)

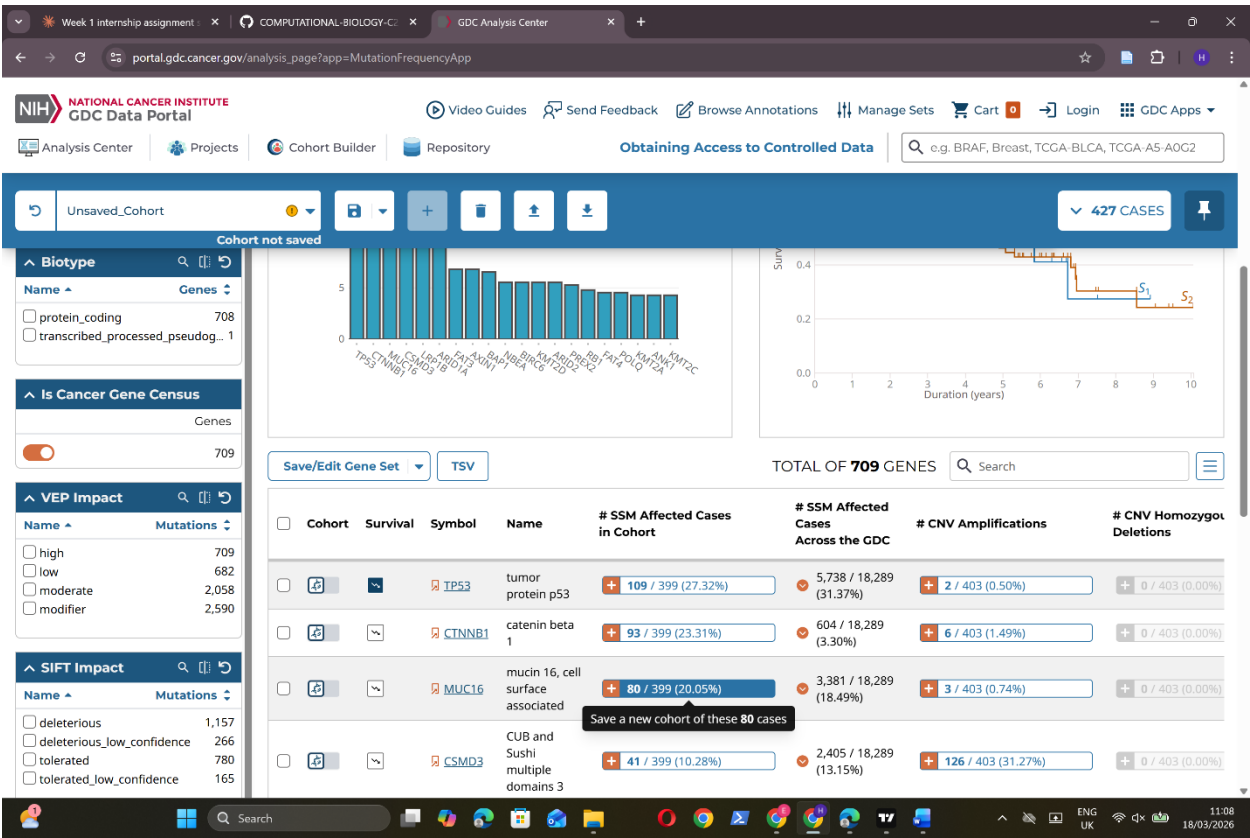
2.a. Mutated genes on display (709)



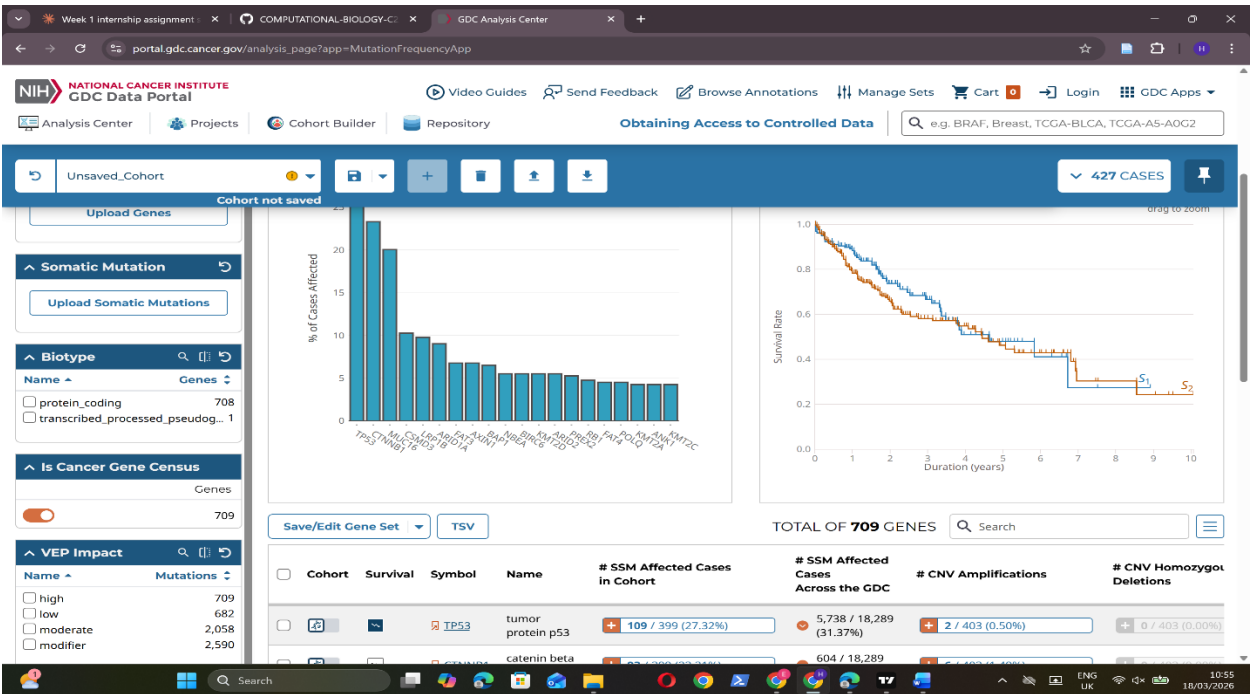
b. Total mutations (3,372)



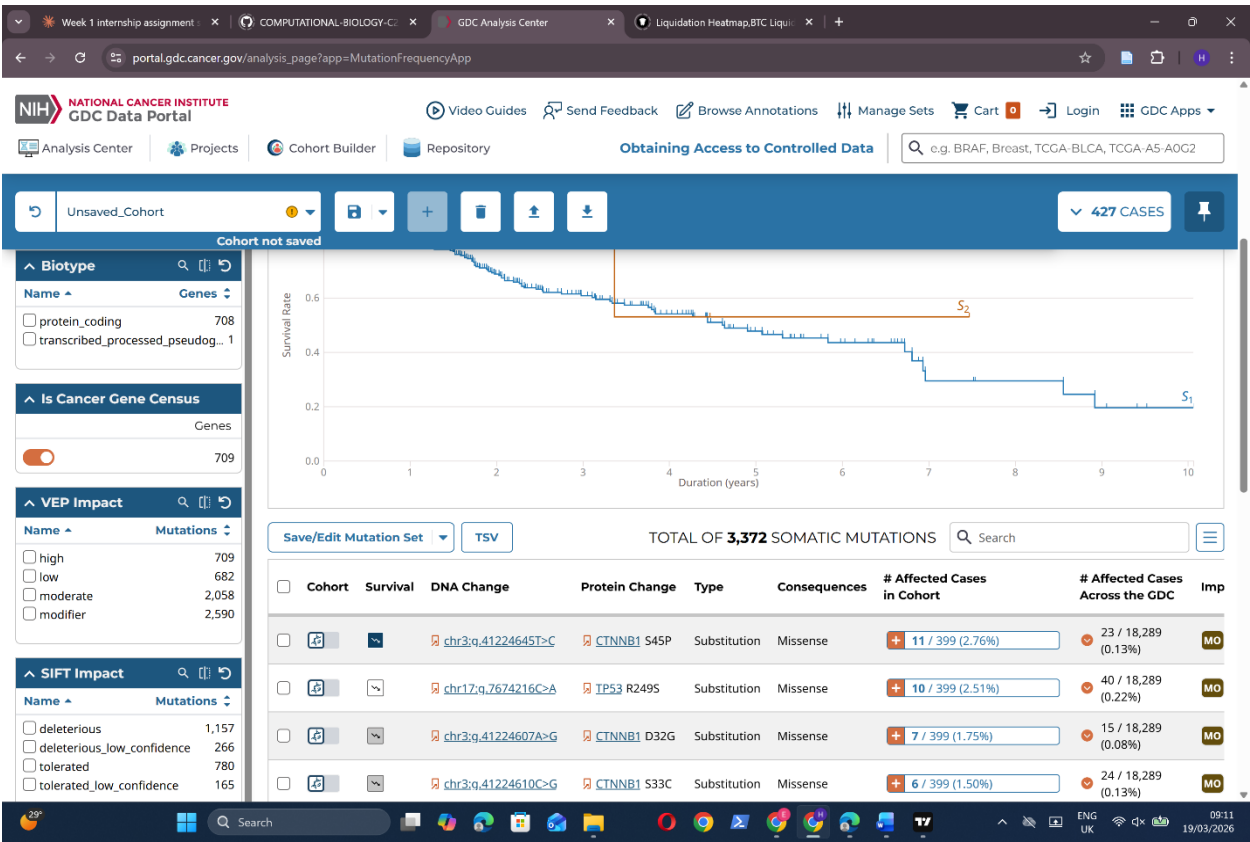
c. Gene with highest number of mutations (TP53 gene)



d. Mutation frequency (27.32%)

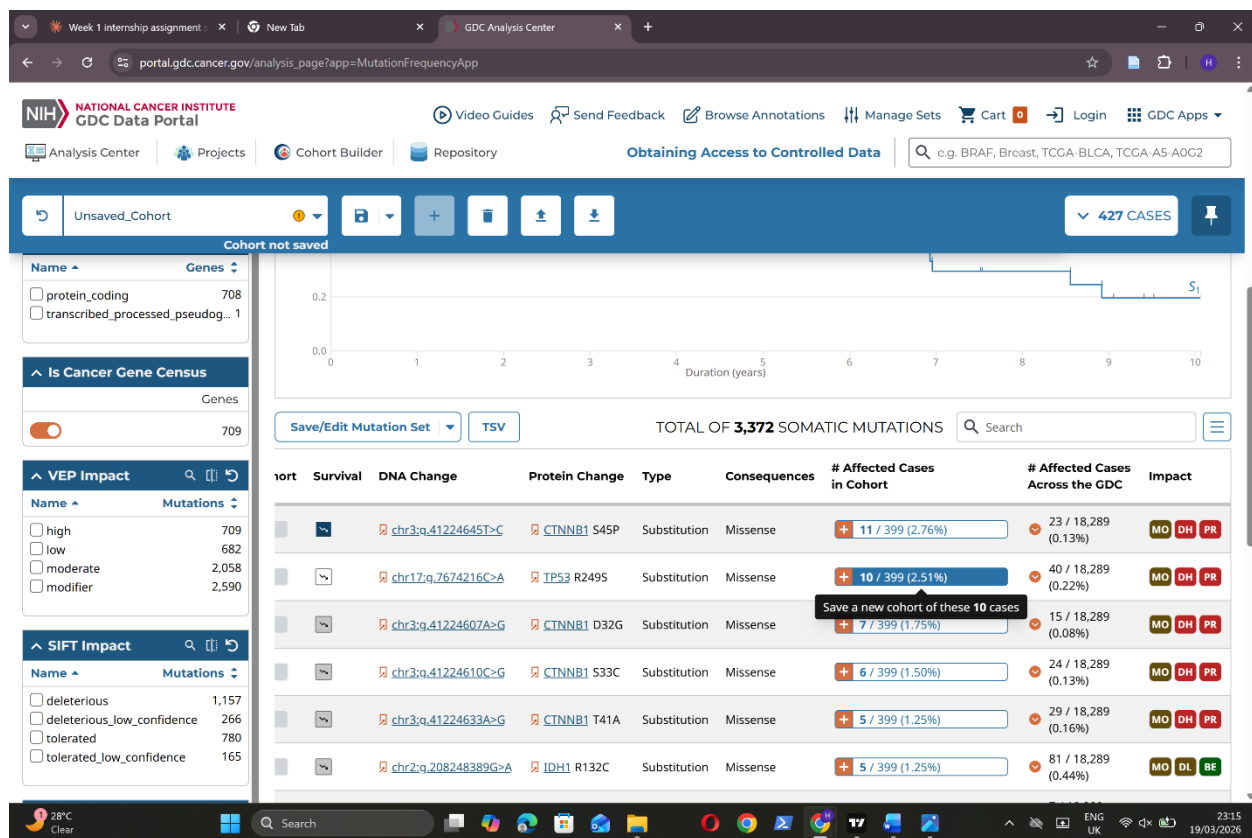


e. Amino acid change Arginine (R) → Serine (S) at position 249.



BIOLOGICAL IMPACT- ADVANCED TASK

2.f. VEP effect: Moderate



g. How this moderate effect affects the protein structure

TP53 is a tumor suppressor gene that codes for the p53 protein, often called the "guardian of the genome." In its normal state, p53 detects DNA damage and either pauses cell division for repair or triggers apoptosis. The R249S mutation disrupts p53's DNA-binding ability, rendering it unable to perform this function and allowing damaged cells to proliferate unchecked, contributing to liver cancer development.

References

- Balasundaram, A., & Doss, C. G. P. (2022). Unraveling the Structural Changes in the DNA-Binding Region of Tumor Protein p53 (TP53) upon Hotspot Mutation p53 Arg248 by Comparative Computational Approach. *International Journal of Molecular Sciences*, 23(24), 15499.
<https://doi.org/10.3390/ijms232415499>